AP CALCULUS AB

Unit 1 • Lesson 1.14 practice

Lesson 1.14 • Connecting Infinite Limits and Vertical Asymptotes

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to 0^-} \frac{1}{x}$ is

(A) $+\infty$

(B) $-\infty$

(C) 0

(D) 1

Question 2

NO CALCULATOR

$\lim_{x \to 2} \frac{3}{(x-2)^2}$ is

(A) $+\infty$

(B) $-\infty$

(C) 3

(D) It does not exist and is not infinite.

Question 3

NO CALCULATOR

Which line is a vertical asymptote of $f(x) = \ln(x-1)$?

(A) $x = 0$

(B) $x = e$

(C) $x = 1$

(D) $y = 1$

Question 4

NO CALCULATOR

$\lim_{x \to \frac{\pi}{2}^+} \tan x$ is

(A) $+\infty$

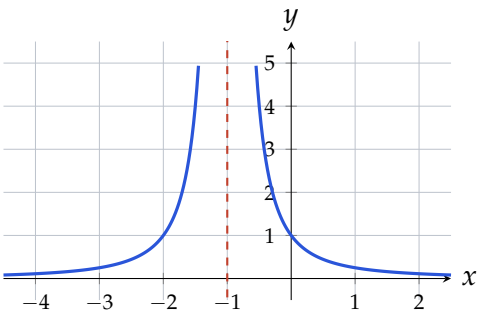
(B) 0

(C) 1

(D) $-\infty$

Question 5 (graph) NO CALCULATOR

The graph of $f(x) = \dfrac{1}{(x+1)^2}$ is shown. Which describes $\lim\limits_{x \to -1} f(x)$?



(A) $+\infty$
(B) $-\infty$
(C) 0
(D) 1

Question 6 (Free response) NO CALCULATOR

Let $f(x) = \dfrac{x+2}{x^2-4x}$.

(a) Find all vertical asymptotes of the graph of f.
(b) For each asymptote, find both one-sided limits. Justify the signs.